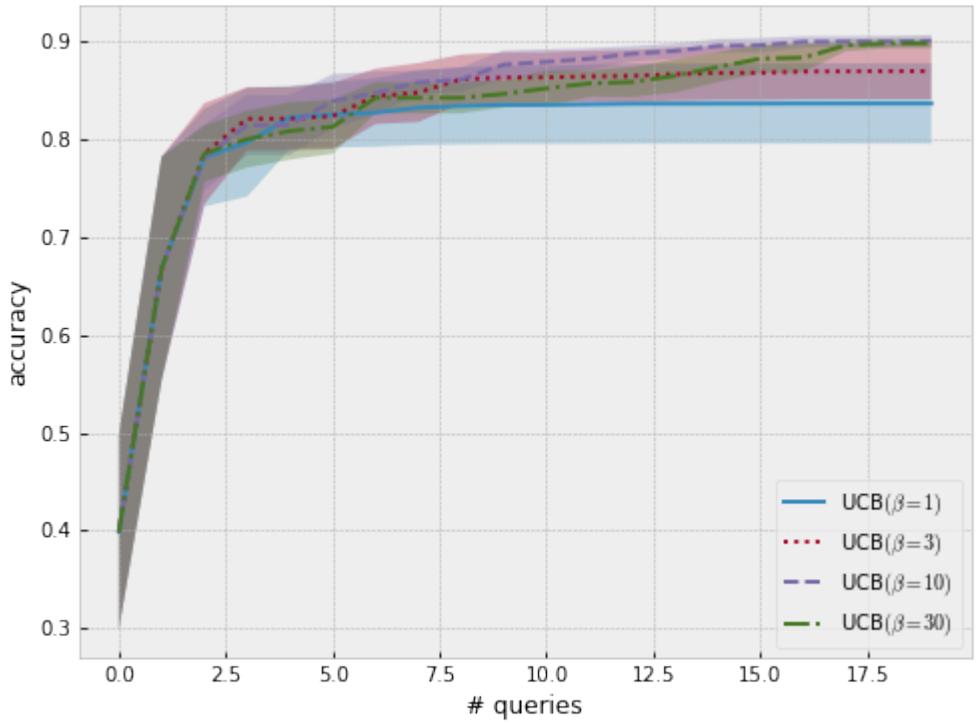


Figure A.11 Progress made by the Upper Confidence Bound policy. The policy is able to escape the loca optimum and get closer to the global optimum.

Figure A.11 shows the optimization performance of the four versions of UCB. We see that when $= 1$, the policy is too exploitative and achieves the worst performance. As the value of the tradeoff parameter increases, performance increases, but when $= 30$, over-exploration causes UCB to be slower at locating an accuracy of 90 percent. Overall, $= 10$ achieves the best performance.

3. Run the adaptive version of UCB (see Exercise 1).

   We modify the BayesOpt loop as follows:

```
num_repeats = 10

start_beta = 3
end_beta = 10

multiplier = (end_beta / start_beta) ** (1 / num_queries)

incumbents = torch.zeros((num_repeats, num_queries))

for trial in range(num_repeats):
    ... ❶

    beta = start_beta

    for i in tqdm(range(num_queries)):
        ... ❷

        policy = botorch.acquisition.analytic.UpperConfidenceBound(
            model, beta=beta
        )

        ... ❸

        beta *= multiplier
```

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